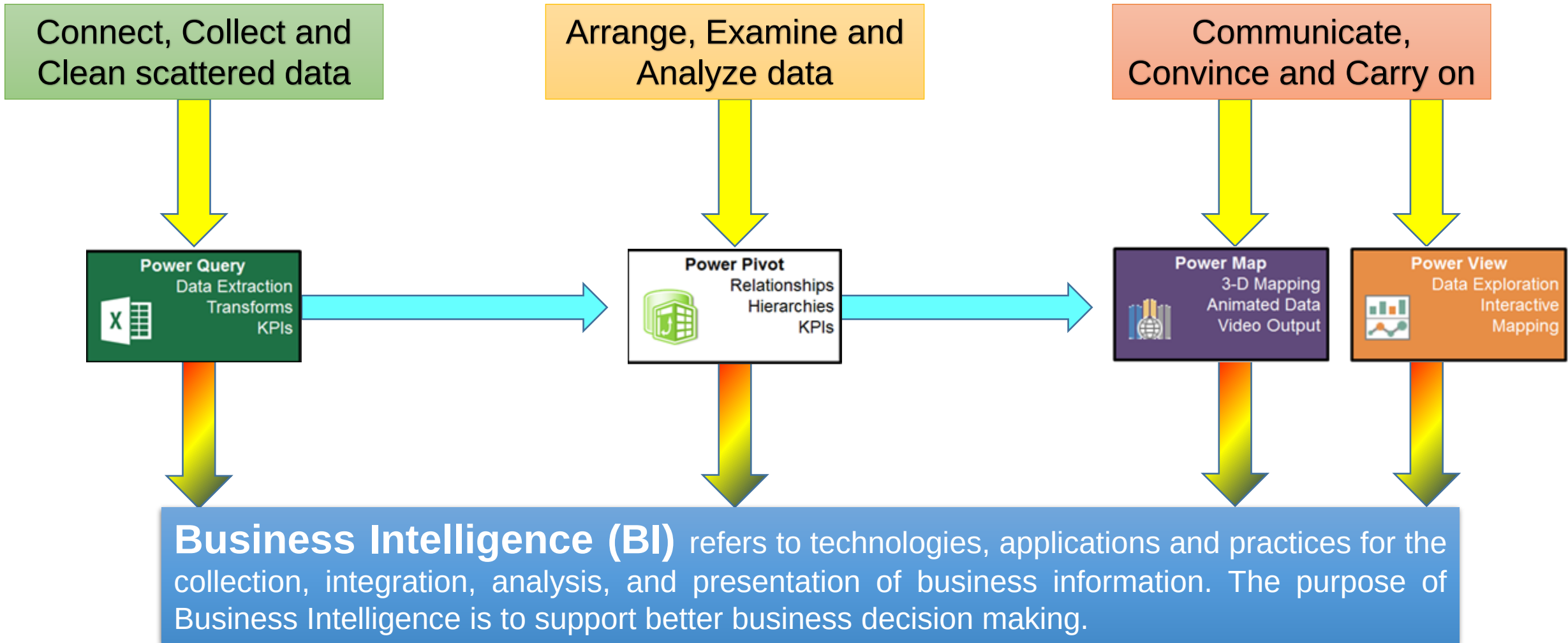


POWER QUERY

Data machine to transform your data

Modern Challenges for everyone

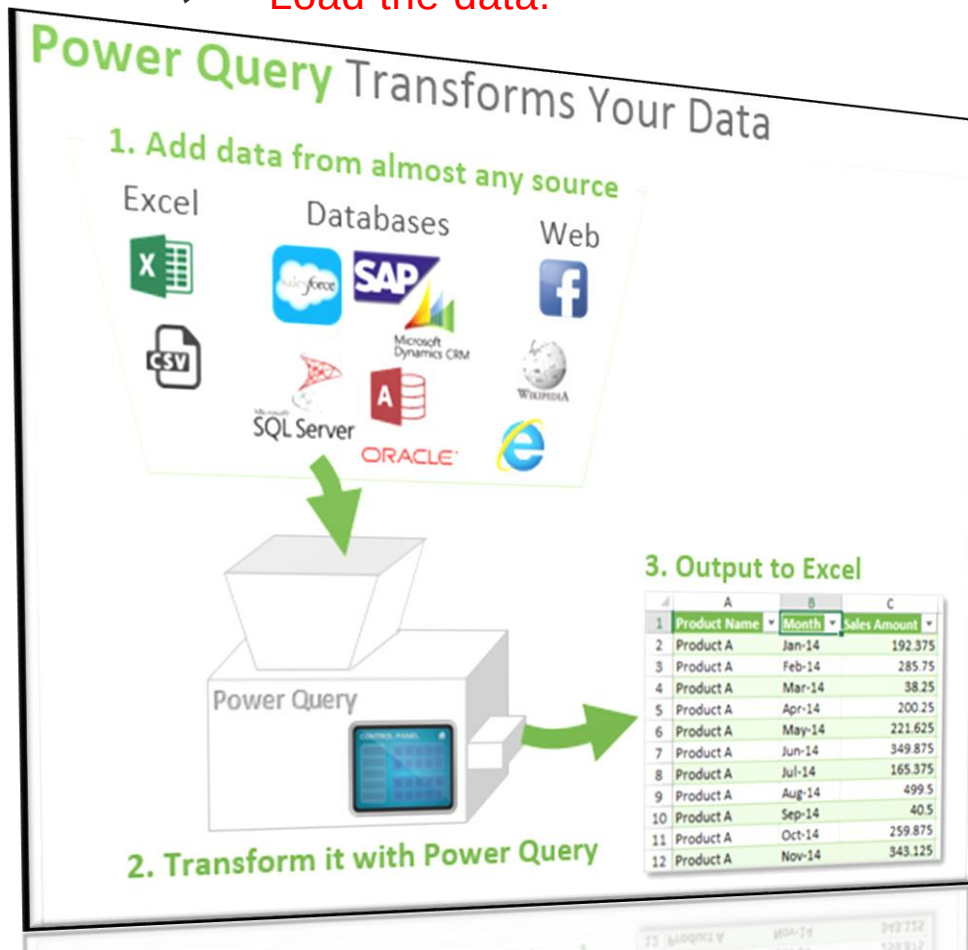


What is Power Query

Power Query is an ETL tool. Its function is to:

- ✓ Extract data from almost every source.
- ✓ Transform data.
- ✓ Load the data.

- Power query is a macro recorder that keeps track of data Extract and Transform steps.
- Define query once and determine where you want to load your data.
- Refresh your query to run it again.



Extract	Transform	Load
• CSV • Text • Excel • Database • Folders • Web • Servers • Marketplace • Share point • Online services • Data feed • Blank query	✓ Data Cleaning. ✓ Data Integration. (Lookups, Index, Match) ✓ Join data. ✓ Merge Tables. ✓ Append Data.	1. Excel Tables. 2. Power Pivot Data Model. 3. Power BI. 4. Connections only.

Import Delimited Files

The screenshot displays the Microsoft Power Query Editor interface. The ribbon at the top includes tabs for File, Home, Transform, Add Column, and View. The Transform tab is active, showing various data manipulation options like Merge Queries, Append Queries, and Replace Values. The main workspace shows a data table with columns: TranDate, Account, Dept, and Sum of Amount. The formula bar at the top displays the query definition: `= Table.TransformColumnTypes("#Promoted Headers",{"TranDate", type date}, {"Account", Int64.Type}, {"Dept", Int64.Type}, {"Sum of`. The 'Query Settings' pane on the right shows the query name 'Session-1-Ex-1-Delimited' and a list of applied steps: Source, Promoted Headers, and Changed Type. A large, stylized text overlay in the center reads 'Welcome to Power Query Environment'.

	TranDate	Account	Dept	Sum of Amount
1	01/12/2009	61510	150	-22.07
2	01/12/2009	61520	150	-151.82
3	01/12/2009	61530	150	-12.4
4	01/12/2009	61540	150	-0.92
5	01/12/2009	61550	150	-61.87
6	01/12/2009	61560	150	-1.6
7	01/12/2009	61570	150	-127.03
8	01/12/2009	62010	150	-283.84
9	01/12/2009	62020	150	-241.45
10	01/12/2009	62099	150	18.41
11	01/12/2009	62510	120	-70.58
12	01/12/2009	62520	120	-73.52
13	01/12/2009	62530	120	2.73
14	01/12/2009	62550	120	-15.5
15	01/12/2009	62560	120	-1.45
16	01/12/2009	63050	150	-10.87
17	01/12/2009	64000	150	180.71
18	01/12/2009	64010	150	1.88
19	01/12/2009	64020	150	0.83
20	01/12/2009	65540	110	-72.39
21	01/12/2009	65550	110	-31.14
22	01/12/2009	65590	110	-6.05
23	01/12/2009	65600	110	-12.03
24	01/12/2009	65610	110	-26.33

Import Excel Data

Excel Tabular Data

- Excel will kick off the process of creating official Excel Table.
- On Confirmation, it will launch Power Query Interface.
- Table Headers will be imported from the Official Excel Table, so there is no Promoted Headers step.

Named Range

- Define a Named Range that covers the data.
- Select the Named Range.
- Create a new query: From Table

Overriding Power Query Defaults

- ✓ Removing Power Query steps.
- ✓ Reconfiguring steps.
- ✓ Re-ordering steps.
- ✓ Deleting steps.
- ✓ Forcing Data types.

Session-1-Ex-3-OverridingDefaults.csv

File Origin	Delimiter	Data Type Detection
1252: Western European (Windows)	Comma	Based on first 200 rows

Column1	Column2	Column3	Column4
Downloaded from MarSys: 2008-02-13T10:13:29			
TranDate	Account	Dept	Amount
1/2/2008	61510	150	(\$26.03)
1/2/2008	61520	150	(\$55.07)
1/2/2008	61530	150	(\$10.60)
1/2/2008	61540	150	(\$0.29)
1/2/2008	61550	150	(\$48.02)
1/2/2008	61560	150	(\$1.35)
1/2/2008	61570	150	(\$77.04)

Load Edit Cancel

fx = Table.TransformColumnTypes("#Promoted Headers",{{"Amount", type number}})

	A ^B _C TranDate	A ^B _C Account	A ^B _C Dept	1.2 Amount
1	1/2/2008	61510	150	-26.03
2	1/2/2008	61520	150	-55.07
3	1/2/2008	61530	150	-10.6
4	1/2/2008	61540	150	-0.29
5	1/2/2008	61550	150	-48.02
6	1/2/2008	61560	150	-1.35
7	1/2/2008	61570	150	-77.04
8	1/2/2008	62010	150	-305.95
9	1/2/2008	62020	150	-95.15
10	1/2/2008	62099	150	8.79
11	1/2/2008	62510	120	-56.74
12	1/2/2008	62520	120	-41.22
13	1/2/2008	62530	120	-3.48
14	1/2/2008	62550	120	-8.97
15	1/2/2008	62560	120	-0.46
16	1/2/2008	63050	150	-7.17
17	1/2/2008	64000	150	186.76

Query Settings

PROPERTIES

Name

Session-1-Ex-3-OverridingDefaults

All Properties

APPLIED STEPS

Source *

Removed Top Rows *

Split Column by Delimiter *

Changed Type *

Promoted Headers *

✕ Changed Type1

Append Operations

- ✓ Import Data files.
- ✓ Append one table to another.
- ✓ Append additional tables.
- ✓ Combine queries with different headers.

Append

☒ Two tables ☐ Three or more tables

Primary table
Session-1-Ex-4A-Jan2008

Table to append to the primary table
Session-1-Ex-4B-Feb2008

Append one table to another.

OK Cancel

Split Column Group By Replace Values Transform

Data Type: Date Use First Row as Headers

Merge Queries Append Queries Combine Files Combine

Append

☒ Two tables ☐ Three or more tables

Table to append
Session-1-Ex-4C-Mar2008

Append additional tables

OK Cancel

As long as the headers of the queries being combined are identical, the second query will just be appended to first query. But what if the columns do not have the same headers?

To fix the issue, Edit the query and Rename the column headers.

	Date	1 ² 3 Account	1 ² 3 Dept	1.2 Amount	1.2 Sales
1	02/01/2008	61510	150	-26.03	null
2	02/01/2008	61520	150	-55.07	null
3	02/01/2008	61530	150	-10.6	null
4	02/01/2008	61540	150	-0.29	null
5	02/01/2008	61550	150	-48.02	null
6	02/01/2008	61560	150	-1.35	null
7	02/01/2008	61570	150	-77.04	null
8	02/01/2008	62010	150	-305.95	null
9	02/01/2008	62020	150	-95.15	null
10	02/01/2008	62099	150	8.79	null
11	02/01/2008	62510	120	-56.74	null

Importing Files from a Folder

- ✓ Extract Files listing.
- ✓ Combine Files.
- ✓ Adding more Files.
- ✓ Taking date from file name.
- ✓ Merging File Properties with Binary Content.

Content	Name	Extension	Date accessed	Date modified	Date created	Attributes
Binary	Feb 2008.csv	.csv	29/08/2015 11:23:24 AM	13/05/2018 11:40:00 PM	29/08/2015 11:23:24 AM	Record
Binary	Jan 2008.csv	.csv	29/08/2015 11:23:20 AM	13/05/2018 11:40:00 PM	29/08/2015 11:23:20 AM	Record
Binary	Mar 2008.csv	.csv	29/08/2015 11:23:22 AM	13/05/2018 11:40:00 PM	29/08/2015 11:23:22 AM	Record

Call files from a folder

	A ^B C	Name	Content
1		Feb 2008.csv	Binary
2		Jan 2008.csv	Binary
3		Mar 2008.csv	Binary
4		Apr 2008.csv	Binary
5		Aug 2008.csv	Binary
6		Dec 2008.csv	Binary
7		Jul 2008.csv	Binary
8		Jun 2008.csv	Binary
9		May 2008.csv	Binary
10		Nov 2008.csv	Binary
11		Oct 2008.csv	Binary
12		Sep 2008.csv	Binary

Remove file extension and change the date

	A ^B C	Name	Content
		Feb 1, 2008	Binary
		Jan 1, 2008	Binary
		Mar 1, 2008	Binary
		Apr 1, 2008	Binary
		Aug 1, 2008	Binary
		Dec 1, 2008	Binary
		Jul 1, 2008	Binary
		Jun 1, 2008	Binary
		May 1, 2008	Binary
		Nov 1, 2008	Binary
		Oct 1, 2008	Binary
		Sep 1, 2008	Binary

Add month end date

	Date	Content
	29/02/2008	Binary
	31/01/2008	Binary
	31/03/2008	Binary
	30/04/2008	Binary
	31/08/2008	Binary
	31/12/2008	Binary
	31/07/2008	Binary
	30/06/2008	Binary
	31/05/2008	Binary
	30/11/2008	Binary
	31/10/2008	Binary
	30/09/2008	Binary

To extract contents of binary files, add a custom column

Custom Column

New column name

Custom

Custom column formula:

= Csv.Document([Content])

Merge files

	Date	ABC 123	Custom
	29/02/2008		Table
	31/01/2008		Table
	31/03/2008		Table
	30/04/2008		Table
	31/08/2008		Table
	31/12/2008		Table
	31/07/2008		Table
	30/06/2008		Table
	31/05/2008		Table
	30/11/2008		Table
	31/10/2008		Table
	30/09/2008		Table

Import files from a folder, convert file name to date and add new column as date

Aggregating data from Workbooks

Generate a list of Excel Workbooks and extract their contents.

Content	Name	Extension	Date accessed	Date modified	Date c
Binary	Denmark.xlsm	.xlsm	20/05/2018 6:51:17 PM	05/05/2018 7:59:00 PM	20/05/
Binary	France.xlsx	.xlsx	20/05/2018 6:51:17 PM	20/05/2018 6:50:56 PM	20/05/
Binary	Germany.xlsx	.xlsx	20/05/2018 6:51:17 PM	05/05/2018 7:58:13 PM	20/05/
Binary	Greece.xlsx	.xlsx	20/05/2018 6:51:17 PM	05/05/2018 7:57:55 PM	20/05/
Binary	Italy.XLSX	.XLSX	20/05/2018 6:51:17 PM	05/05/2018 7:57:33 PM	20/05/
Binary	Netherland.xlsx	.xlsx	20/05/2018 6:51:17 PM	05/05/2018 7:56:58 PM	20/05/
Binary	Norway.xlsx	.xlsx	20/05/2018 6:51:17 PM	05/05/2018 7:56:35 PM	20/05/
Binary	Spain.xlsx	.xlsx	20/05/2018 6:51:17 PM	05/05/2018 7:55:51 PM	20/05/
Binary	Test File.txt	.txt	20/05/2018 6:51:17 PM	05/05/2018 8:09:15 PM	20/05/
Binary	United Kingdom.xlsx	.xlsx	20/05/2018 6:51:17 PM	05/05/2018 7:59:26 PM	20/05/

=Excel.Workbook([Content])

New column name

Custom

Custom column formula:

=Excel.Workbook([Content])

Available columns:

Content

Name

ABC Name	ABC Name.1	ABC Data	ABC Item	ABC Kind	ABC Hidden
Denmark.xlsm	Denmark	Table	Denmark	Sheet	FALSE
France.xlsx	France	Table	France	Sheet	FALSE
France.xlsx	AdditionalData	Table	AdditionalData	Sheet	TRUE
France.xlsx	TFrance	Table	TFrance	Table	FALSE
France.xlsx	FranceRange	Table	AdditionalData!FranceRange	DefinedName	FALSE
France.xlsx	FranceRange1	Table	FranceRange	DefinedName	FALSE
Germany.xlsx	Germany	Table	Germany	Sheet	FALSE
Greece.xlsx	Greece	Table	Greece	Sheet	FALSE
Italy.XLSX	Italy	Table	Italy	Sheet	FALSE
Netherland.xlsx	Netherland	Table	Netherland	Sheet	FALSE
Norway.xlsx	Norway	Table	Norway	Sheet	FALSE
Spain.xlsx	Spain	Table	Spain	Sheet	FALSE
United Kingdom.xlsx	United Kingdom	Table	United Kingdom	Sheet	FALSE

ABC Country	Date	ABC Products	123 Sales
Denmark	01/01/2018	Bedding	570251
Denmark	01/01/2018	Blankets	105720
Denmark	01/01/2018	Towels	498941
Denmark	01/01/2018	Carpets	555514
Denmark	01/01/2018	Hosiery	275639
Denmark	01/01/2018	Curtains	435997
Denmark	01/01/2018	Rugs	495549
Denmark	01/01/2018	Zippers	252554
Denmark	01/01/2018	Bags	351188
Denmark	01/01/2018	Canopies	230486
Denmark	02/01/2018	Bedding	432034
Denmark	02/01/2018	Blankets	476580

APPLIED STEPS

- Source
- Lowercased Text
- Filtered Rows
- Removed Other Columns
- Added Custom
- Expanded Custom
- Removed Columns
- Filtered Rows1
- Removed Other Columns1
- Split Column by Delimiter
- Changed Type
- Renamed Columns
- Removed Columns1
- Expanded Data
- Renamed Columns1
- Filtered Rows2
- Changed Type1

Unpivoting Data

- ✓ Your data does not always arrive in a useful state. People will create reports using pivot tables, convert it to range and share with colleagues.
- ✓ Transposing the data will change the look, it does not truly convert data into format that Pivot Tables are ready to consume.

ABC 123	Name	ABC 123	Column1	ABC 123	Column2	ABC 123	Column3	ABC 123	Column4	ABC 123	Column5	ABC 123	Column6	ABC 123	Column7
1	Malaysia		null		Sales per Product		null		null		null		null		null
2	Malaysia		Date		Bedding		Blankets		Towels		Carpets		Hosiery		Curtains
3	Malaysia		01/01/2018		150051		397159		127904		295461		127575		380178
4	Malaysia		02/01/2018		434141		376877		217775		233546		122675		245249
5	Malaysia		03/01/2018		287396		138323		335427		361081		260378		323810
6	Malaysia		04/01/2018		294467		null		311834		185861		325256		266208
7	Malaysia		05/01/2018		154286		316482		325624		402662		154669		256038
8	Malaysia		06/01/2018		158502		389935		181178		220783		465720		234903
9	Malaysia		07/01/2018		332078		410750		331220		440997		306869		299158
10	Malaysia		08/01/2018		257432		448956		336501		177239		168108		250652
11	Malaysia		09/01/2018		247024		211326		393620		150157		120080		147773
12	Malaysia		10/01/2018		129030		254949		223757		310249		275504		177225

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Country	Date	Products	Sales											
2	Malaysia	01/01/2018	Bedding	150051											
3	Malaysia	01/01/2018	Blankets	397159											
4	Malaysia	01/01/2018	Towels	127904											
5	Malaysia	01/01/2018	Carpets	295461											
6	Malaysia	01/01/2018	Hosiery	127575											
7	Malaysia	01/01/2018	Curtains	380178											
8	Malaysia	01/01/2018	Rugs	264084											
9	Malaysia	01/01/2018	Zippers	444024											
10	Malaysia	01/01/2018	Bags	487801											
11	Malaysia	01/01/2018	Canopies	228058											
12	Malaysia	02/01/2018	Bedding	434141											
13	Malaysia	02/01/2018	Blankets	376877											
14	Malaysia	02/01/2018	Towels	217775											
15	Malaysia	02/01/2018	Carpets	233546											
16	Malaysia	02/01/2018	Hosiery	122675											

APPLIED STEPS

- Source ✖
- Removed Other Columns ✖
- Added Custom ✖
- Expanded Custom ✖
- Removed Other Columns1 ✖
- Expanded Data ✖
- Removed Top Rows ✖
- Promoted Headers ✖
- Removed Columns ✖
- Filtered Rows ✖
- Renamed Columns ✖
- Unpivoted Other Columns ✖
- Renamed Columns1 ✖
- ✖ Changed Type

Importing a Non-delimited Text File

Importing and cleaning non-delimited text files is the biggest challenge. You will face a number of issues with non-delimited text files. Such as non-printing characters, repeating row headers, inconsistent alignment, duplicate spaces, etc., etc. Use Trim() and Clean() functions.

A ^B C Column1						
XYZ Company Ltd.			Pg 1			
Detailed General Ledger with Net Change for Aug 2004 to Mar 2009						
Dept xxx - Restaurant Costs						
Account			Begin	Net	Final	
No.	Description of Account	Balance		Change	Balance	
Jan 2006						
123-03	Purchases		0.00	214,861.77	214,861.77	
	Tran Date	Tran Amount	Source	Reference Information	Vendor Name	
	01/02/2006	4,429.36	GJ0001	Food Purchase-North Douglas Distributors		
	01/02/2006	10,450.96	GJ0002	Liquor Purchase-Liquor Distribution Branch		
	01/02/2006	128.25	GJ0003	Supply Purchase-Avis & Davis Distributors Inc		
	01/04/2006	4,895.61	GJ0004	Food Purchase-North Douglas Distributors		
	01/04/2006	11,551.06	GJ0005	Liquor Purchase-Liquor Distribution Branch		
	01/04/2006	141.75	GJ0006	Supply Purchase-Avis & Davis Distributors Inc		
	01/06/2006	4,662.48	GJ0007	Food Purchase-North Douglas Distributors		
	01/06/2006	11,001.01	GJ0008	Liquor Purchase-Liquor Distribution Branch		
	01/06/2006	135.00	GJ0009	Supply Purchase-Avis & Davis Distributors Inc		
	01/09/2006	3,276.19	GJ0010	Food Purchase-North Douglas Distributors		
	01/09/2006	12,773.43	GJ0011	Liquor Purchase-Liquor Distribution Branch		
	01/09/2006	109.92	GJ0012	Supply Purchase-Avis & Davis Distributors Inc		

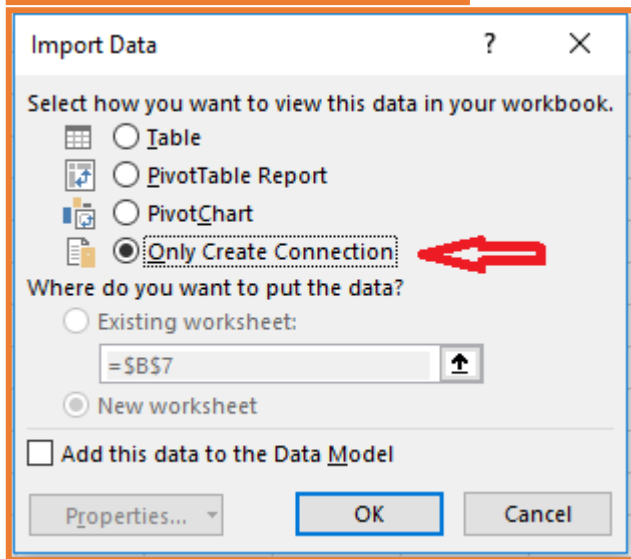
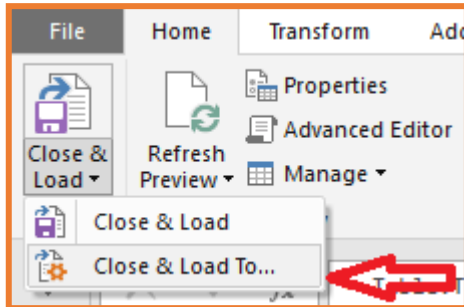
Date	1.2 Amount	Source	Category	Vendor
03/03/2006	5523.92	GJ0079	Food Purchase	North Douglas Distributors
03/03/2006	15222.14	GJ0080	Liquor Purchase	Liquor Distribution Branch
03/03/2006	149.81	GJ0081	Supply Purchase	Avis & Davis Distributors Inc
01/03/2006	157.3	GJ0078	Supply Purchase	Avis & Davis Distributors Inc
27/02/2006	142.32	GJ0075	Supply Purchase	Avis & Davis Distributors Inc
01/03/2006	5800.12	GJ0076	Food Purchase	North Douglas Distributors
01/03/2006	15983.24	GJ0077	Liquor Purchase	Liquor Distribution Branch
08/03/2006	16624.94	GJ0086	Liquor Purchase	Liquor Distribution Branch
08/03/2006	282.65	GJ0087	Supply Purchase	Avis & Davis Distributors Inc
10/03/2006	8557.19	GJ0088	Food Purchase	North Douglas Distributors
08/03/2006	8985.05	GJ0085	Food Purchase	North Douglas Distributors
06/03/2006	8129.33	GJ0082	Food Purchase	North Douglas Distributors
06/03/2006	15041.62	GJ0083	Liquor Purchase	Liquor Distribution Branch
06/03/2006	255.73	GJ0084	Supply Purchase	Avis & Davis Distributors Inc
27/02/2006	14461.03	GJ0074	Liquor Purchase	Liquor Distribution Branch
20/02/2006	5952.31	GJ0064	Food Purchase	North Douglas Distributors
20/02/2006	15983.25	GJ0065	Liquor Purchase	Liquor Distribution Branch
20/02/2006	164.78	GJ0066	Supply Purchase	Avis & Davis Distributors Inc

- Source *
- Removed Top Rows *
- Trimmed Text
- Cleaned Text
- Split Column by Position *
- Promoted Headers *
- Changed Type with Locale *
- Removed Errors
- Filtered Rows *
- Renamed Columns
- Changed Type
- Removed Errors1
- Filtered Rows1 *
- Sorted Rows
- Removed Columns
- Merged Columns *
- Split Column by Delimiter *
- Changed Type1

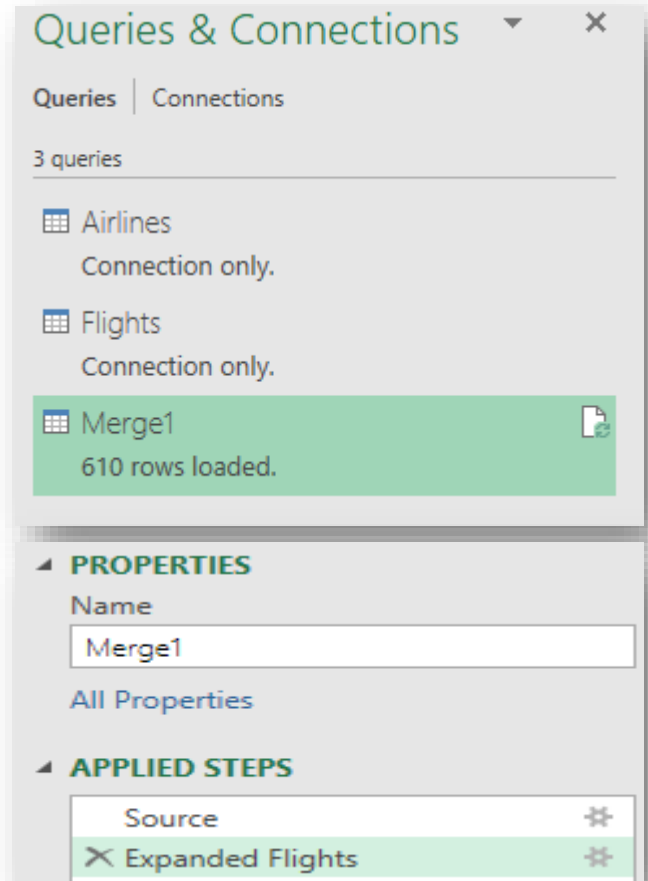
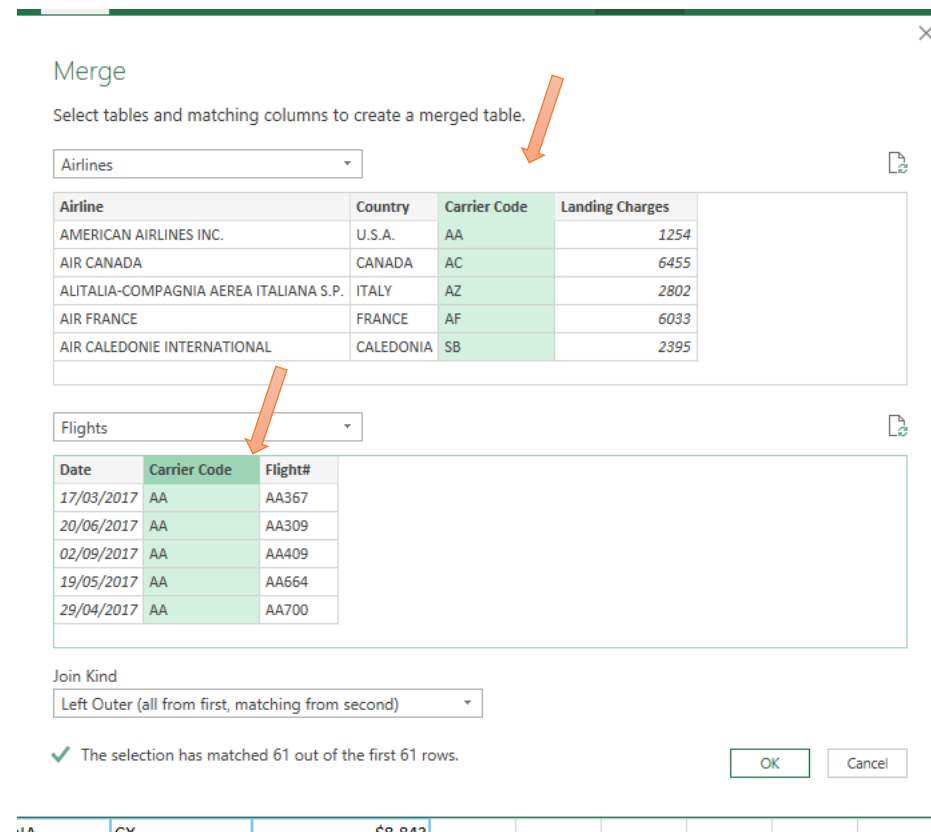
Merging Tables and Queries

A classic problem that Excel users is facing is how to aggregate data from two separate tables into one table to serve a pivot table. The route was to use VLOOKUP() or INDEX(MATCH()). Power Query has a better solution.

Create Pointer Queries



Performing the Merge



Transposing and Unpivoting Complex Data

While unpivoting is extremely common, on occasion you may need to transpose and flip the data so that data can be filled down. There is no right or left fill in Power Query. Create a **Named Range** of the data and pull it in Power Query using **From Table/Range**.

	A ^B _C Class	A ^B _C Account	A ^B _C Month	A ^B _C Measure	1.2 Amount
1	Telecom Revenue	IDD	January	Actual	789451
2	Telecom Revenue	IDD	January	Budget	127132
3	Telecom Revenue	IDD	January	Variance	662319
4	Telecom Revenue	IDD	February	Actual	640168
5	Telecom Revenue	IDD	February	Budget	410197
6	Telecom Revenue	IDD	February	Variance	229971
7	Telecom Revenue	IDD	March	Actual	780826
8	Telecom Revenue	IDD	March	Budget	165927
9	Telecom Revenue	IDD	March	Variance	614899
10	Telecom Revenue	IDD	April	Actual	703288
11	Telecom Revenue	IDD	April	Budget	407057
12	Telecom Revenue	IDD	April	Variance	296231
13	Telecom Revenue	IDD	May	Actual	669419
14	Telecom Revenue	IDD	May	Budget	289013

APPLIED STEPS	
Source	
Changed Type	
Filled Down	
Transposed Table	
Filled Down1	
Merged Columns	✳
Transposed Table1	
Promoted Headers	✳
Changed Type1	
Renamed Columns	
Filtered Rows	✳
Unpivoted Other Columns	
Split Column by Delimiter	✳
Changed Type2	
Renamed Columns1	
✕ Changed Type3	

Grouping and Summarizing Data

When you are working with large amount of data, you need to group and summarize records at the sources level. This will help you to avoid importing unnecessary data for analysis and reports.

	Date	Country	No of Items	Total Sales Per Country
1	01/01/2016	Australia	8	1490672
2	01/01/2016	Canada	8	1252198
3	01/01/2016	England	8	1498163
4	01/01/2016	Japan	8	1249455
5	01/01/2016	Norway	8	1486603
6	01/01/2016	Portugal	8	1568224
7	01/01/2016	Spain	8	1113601
8	01/01/2016	Sweden	8	1612891
9	01/01/2016	USA	8	1232421

Date	Country	No of Items	Total Sales Per Country	Brands	Country.1	Amount	Top Sales Percentage
01/01/2016	Australia	8	1490672	Zara	Australia	281210	19.00%
01/01/2016	Canada	8	1252198	Adidas	Canada	269159	21.00%
01/01/2016	England	8	1498163	Clarks	England	262664	18.00%
01/01/2016	Japan	8	1249455	Adidas	Japan	265801	21.00%
01/01/2016	Norway	8	1486603	Fendi	Norway	291983	20.00%
01/01/2016	Portugal	8	1568224	Polo	Portugal	272967	17.00%
01/01/2016	Spain	8	1113601	Zara	Spain	279900	25.00%
01/01/2016	Sweden	8	1612891	Topman	Sweden	280843	17.00%
01/01/2016	USA	8	1232421	Zara	USA	238015	19.00%
02/01/2016	Australia	8	1329669	Polo	Australia	276521	21.00%
02/01/2016	Canada	8	1486925	Topman	Canada	295601	20.00%
02/01/2016	England	8	1750719	Topman	England	298404	17.00%

APPLIED STEPS

Source
Changed Type
Changed Type1
X Grouped Rows

APPLIED STEPS

Source
Changed Type
Changed Type1
Grouped Rows
Added Custom
Expanded Top Sales
Changed Type2
Added Custom1
Rounded Off
Changed Type3
X Removed Columns

Grouping

Total sales each day by country with number of items that made-up Total sales.

Summarizing

The largest Brand's proportion and Percentage in Total Daily Sales.

=Table.Max([Details],"Amount")

Transposing Stacked Data

Another complicated transformation scenario is one where the details for a transaction are stacked vertically on top of each other with blank rows separating the transactions. Use **Index Column** and **Modulo Column** trick to solve this problem.

Transactions
42303
City of Vancouver
555164
213
42304
Earl's Restauarant
657607
404
42305
Best Buy
741012
396
42306
Kabul Farms
731581
261

ABC 123	Transactions	1.2 Index	1.2 Modulo
	26/10/2015 12:00:00 AM	0	0
	City of Vancouver	1	1
	555164	2	2
	213	3	3
	null	4	4
	null	5	5
	27/10/2015 12:00:00 AM	6	0
	Earl's Restauarant	7	1
	657607	8	2
	404	9	3
	null	10	4
	null	11	5
	28/10/2015 12:00:00 AM	12	0
	Best Buy	13	1
	741012	14	2
	396	15	3
	null	16	4
	null	17	5
	29/10/2015 12:00:00 AM	18	0
	Kabul Farms	19	1
	731581	20	2
	261	21	3

Date	Vendor	TransactionID	Amount
26/10/2015	City of Vancouver	555164	213
27/10/2015	Earl's Restauarant	657607	404
28/10/2015	Best Buy	741012	396
29/10/2015	Kabul Farms	731581	261
30/10/2015	Quick Chick	452082	162
31/10/2015	Chapters	681713	453
01/11/2015	Simons	776295	360
02/11/2015	Bamyan Kabob	575846	210
03/11/2015	Walmart	334012	28
04/11/2015	Costco	927174	198
05/11/2015	ESSO	499227	90
06/11/2015	My Electronics	857274	321
07/11/2015	Speedy	550438	231
08/11/2015	City of Mississauga	408097	185
09/11/2015	Earl's Restauarant	849354	208
10/11/2015	Best Buy	369732	488
11/11/2015	Kabul Farms	717447	90
12/11/2015	Quick Chick	947405	402
13/11/2015	Chapters	872134	146
14/11/2015	Simons	717903	417
15/11/2015	Bamyan Kabob	265777	120

APPLIED STEPS

Source

Changed Type

Added Index

X Inserted Modulo

Pivoted Column

Filled Up

Filtered Rows

Removed Other Columns

Changed Type1

Renamed Columns

Conditional Logic in Power Query

Sometime we need to perform some logic in a column. Power Query's **try** statement is equivalent of **IFERROR()** in Excel. To get time from a specific field, use **Time.From** function. To test some logic, use **if then else** statement.

Date: Time Sheet Report
From: 3/1/2016
To: 3/14/2016

Work Date	Out	Reg Hrs	OT Hrs	Misc
Contingent Worker: Fahad Raza				
3/3/2016	6:00 PM	8.50	0.00	0.00
3/4/2016	6:00 PM	8.00	0.00	0.00
3/5/2016	6:00 PM	8.50	0.00	0.00
3/6/2016	6:00 PM	8.50	0.00	0.00
3/7/2016	3:30 PM	6.50	0.00	0.00
3/10/2016	6:00 PM	8.50	0.00	0.00
3/11/2016	6:00 PM	8.50	0.00	0.00
3/12/2016	6:00 PM	6.00	0.00	0.00
3/13/2016	6:00 PM	8.50	0.00	0.00
3/14/2016	6:00 PM	8.50	0.00	0.00
Number of Record(s): 10 80.00 0.00 0.00				
Contingent Worker: Dilshard Harris				
3/3/2016	6:00 PM	9.50	0.00	0.00
3/4/2016	6:00 PM	8.00	0.00	0.00
3/5/2016	6:00 PM	8.50	0.00	0.00
3/6/2016	5:00 PM	7.50	0.00	0.00
3/7/2016	3:00 PM	6.00	0.00	0.00
3/10/2016	6:00 PM	8.50	0.00	0.00
3/11/2016	6:00 PM	8.50	0.00	0.00
3/12/2016	6:00 PM	6.00	0.00	0.00
3/13/2016	6:00 PM	8.50	0.00	0.00
3/14/2016	6:00 PM	8.50	0.00	0.00
Number of Record(s): 10 79.50 0.00 0.00				

Work Date	Out	Reg Hrs	Employee
03/03/2016	6:00:00 PM	8.5	Fahad Raza
04/03/2016	6:00:00 PM	8	Fahad Raza
05/03/2016	6:00:00 PM	8.5	Fahad Raza
06/03/2016	6:00:00 PM	8.5	Fahad Raza
07/03/2016	3:30:00 PM	6.5	Fahad Raza
10/03/2016	6:00:00 PM	8.5	Fahad Raza
11/03/2016	6:00:00 PM	8.5	Fahad Raza
12/03/2016	6:00:00 PM	6	Fahad Raza
13/03/2016	6:00:00 PM	8.5	Fahad Raza
14/03/2016	6:00:00 PM	8.5	Fahad Raza
03/03/2016	6:00:00 PM	9.5	Dilshard Harris
04/03/2016	6:00:00 PM	8	Dilshard Harris
05/03/2016	6:00:00 PM	8.5	Dilshard Harris
06/03/2016	5:00:00 PM	7.5	Dilshard Harris
07/03/2016	3:00:00 PM	6	Dilshard Harris
10/03/2016	6:00:00 PM	8.5	Dilshard Harris
11/03/2016	6:00:00 PM	8.5	Dilshard Harris
12/03/2016	6:00:00 PM	6	Dilshard Harris
13/03/2016	6:00:00 PM	8.5	Dilshard Harris
14/03/2016	6:00:00 PM	8.5	Dilshard Harris
03/03/2016	6:00:00 PM	9	Rayyan Qaisar

Syntax

Time.From(value as any)

Time.From([Out])

try <operation> otherwise <alternate result>

try Time.From([Out]) otherwise null

APPLIED STEPS

Source

Changed Type

Removed Top Rows

Promoted Headers

Changed Type1

Added Custom

Added Custom1

Filled Down

Filtered Rows

✕ Changed Type2

Importing Web Data

One of the really interesting use cases for Power Query is when leveraging it to pull data relevant to your business from the web.

Pages with Tables

- Use Anonymous authentication.
- https://en.wikipedia.org/wiki/List_of_countries_and_dependencies_by_population

Web hosted files

- It will take you directly to Power Query environment.
- <https://www.domain.com/robots.txt>

Pages without Tables

- Use HTML source code
- Not in the scope of this course

Practice Problem [Index, Modulo column and If statement]

Practice how to use Index Column, Modulo Column and if statement in Power Query.

Commercial Loan Report		
Crux Bank		
For the year 2017		
Customer #	City	State
75	Toronto	ON
Loan #	Loan Rate	Loan Balance
25	2.10%	121,876
Approved Date	Maturity Date	Branch Code
01/01/2017	01/04/2017	12967
Customer #	City	State
76	Mississauga	ON
Loan #	Loan Rate	Loan Balance
26	2.20%	112,701
Approved Date	Maturity Date	Branch Code
06/01/2017	06/04/2017	12710
Customer #	City	State
77	Ottawa	ON
Loan #	Loan Rate	Loan Balance
27	2.75%	114,390

Customer #	City	State	Loan #	Loan Rate	Loan Balance	Approved Date	Maturity Date	Branch Code
75	Toronto	ON	25	0.021	121876	01/01/2017	01/04/2017	12967
76	Mississauga	ON	26	0.022	112701	06/01/2017	06/04/2017	12710
77	Ottawa	ON	27	0.0275	114390	11/01/2017	11/04/2017	11809
78	Hamilton	ON	28	0.021	61597	16/01/2017	16/04/2017	13735
79	Kingston	ON	29	0.0271	128765	21/01/2017	21/04/2017	10233
80	Windsor	ON	30	0.0238	96934	26/01/2017	26/04/2017	14541
81	Sudbury	ON	31	0.031	87975	31/01/2017	01/05/2017	13641
82	London	ON	32	0.0332	69981	05/02/2017	06/05/2017	13393
83	Guelph	ON	33	0.0336	87323	10/02/2017	11/05/2017	14227
84	Barrie	ON	34	0.0258	80955	15/02/2017	16/05/2017	11634
85	Brantford	ON	35	0.0245	71379	20/02/2017	21/05/2017	13226
86	Peterborough	ON	36	0.0198	67674	25/02/2017	26/05/2017	13811
87	Kitchener	ON	37	0.0285	60151	02/03/2017	31/05/2017	12102
88	North Bay	ON	38	0.0222	50598	07/03/2017	05/06/2017	10457
89	Belleville	ON	39	0.0325	117746	12/03/2017	10/06/2017	14872
90	Brampton	ON	40	0.031	84882	17/03/2017	15/06/2017	12154
91	Oshawa	ON	41	0.033	91654	22/03/2017	20/06/2017	11840
92	Thunder Bay	ON	42	0.0301	51433	27/03/2017	25/06/2017	14148
93	St Catharines	ON	43	0.0321	108672	01/04/2017	30/06/2017	11947
94	Orillia	ON	44	0.0226	111579	06/04/2017	05/07/2017	11535

APPLIED STEPS	
Source	
Changed Type	
Removed Top Rows	✖
Added Index	✖
Inserted Modulo	✖
Added Custom	✖
Added Custom1	✖
Added Custom2	✖
Added Custom3	✖
Added Custom4	✖
Added Custom5	✖
Added Custom6	✖
Added Custom7	✖
Added Custom8	✖
Filled Down	
Filtered Rows	✖
Removed Other Columns	✖
✖ Changed Type1	

Thank you